

Modern Web Standards

BUILDING RESILIENT SITES



Modern Web Standards...

Shared rules that keep the web open, interoperable, and robust across devices and browsers. They cover HTML, CSS, JavaScript, accessibility, performance, and security practices. This means we:

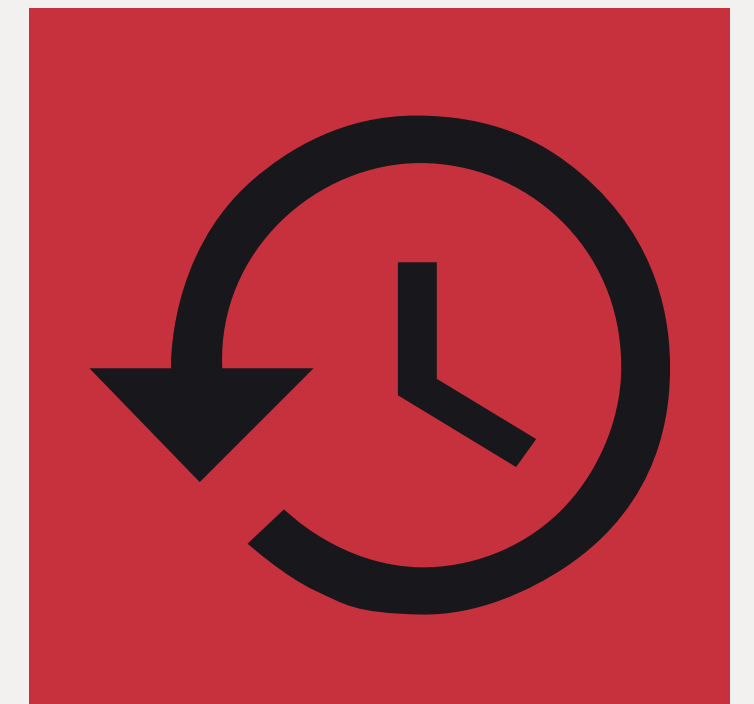
- Use well-structured, semantic HTML to give content clear meaning.
- Separate content, presentation, and behavior so sites are maintainable.
- Aim for cross browser and device compatibility instead of relying on proprietary or fragile features.

Standards evolve, so it's a commitment to stay aligned with current best practices.



The Historical Evolution

- The W3C was founded in 1994 to guide the web through open, shared standards
- In the 1990s, browser competition created incompatible implementations and fragile websites
- Designers often used HTML tables for layout because CSS support was still immature
- In 1998, the Web Standards Project (WaSP) began pushing browsers to support standards properly
- Better support for HTML, CSS, and JavaScript helped the industry move toward semantic, maintainable, standards-based web design



Why it matters

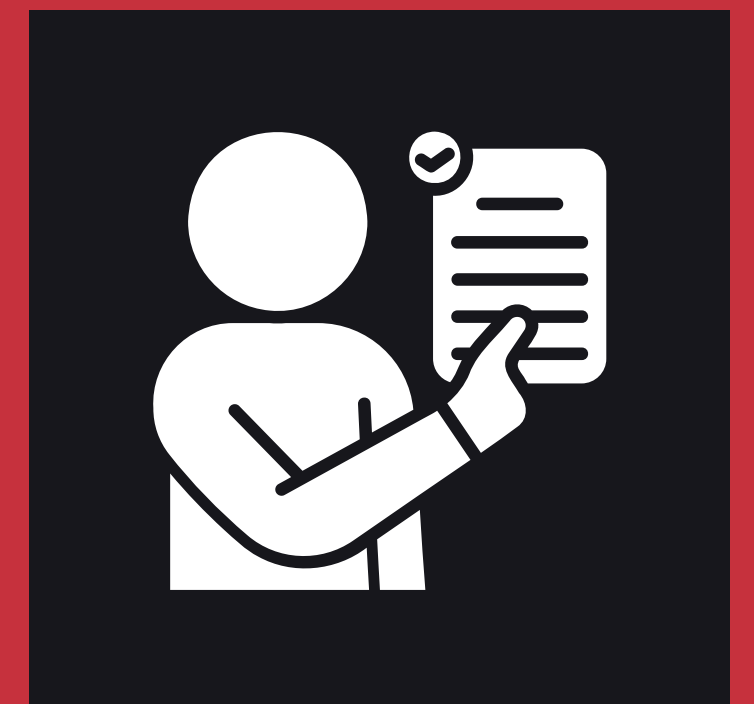
Web standards reduces technical debt, improves performance, and helps site to works well for more people, in more places, for longer.

- Makes sites easier to update, extend, and integrate with other tools.
- Visitors get faster load times, better accessibility, and fewer glitches.
- We avoid unnecessary lock-in to proprietary technologies (where an open standard exists).



Core principles

- **Progressive enhancement:** Start with a solid, accessible core that works in basic conditions, then layer on enhancements for capable browsers and devices.
- **Semantic structure:** Use HTML elements for their intended purpose so content is meaningful to both people and machines (browsers, search engines, assistive tech).
- **Responsive design:** Design layouts, typography, and media that adapt gracefully to different screen sizes and orientations.
- **Performance and resilience:** Favour lightweight assets, efficient code, and graceful fallbacks over fragile, heavy solutions.
- **Security and privacy:** Use secure protocols (HTTPS), avoid unsafe patterns, and respect user data.



How website owners apply this

Website owners don't need to know every technical detail, but they can still make decisions that reinforce standards-based practice:

- Prefer solutions that work in a regular browser without special plugins or proprietary viewers.
- Avoid demanding features that rely on “magic” if they create performance or accessibility problems.
- Keep content structured with meaningful headings, lists, and labels rather than styling alone
- Encourage use of responsive layouts and media that work well on mobile, tablet, and desktop.
- Periodically review the site for outdated technologies (unsupported APIs, insecure embeds, obsolete formats) and plan to replace them



References

- W3C – Web Standards and Guidelines: overview of the core standards that shape the web
- MDN Web Docs – Web Standards: practical, developer-friendly. documentation on HTML, CSS, and JavaScript
- Web.dev (by Chrome team) – guides on modern best practices for performance, accessibility, and PWA features
- WHATWG – HTML Living Standard: the continuously updated definition of HTML
- The W3C standards overview — current framing of web standards.w3
The coming of web standards — history from the W3C community.w3

